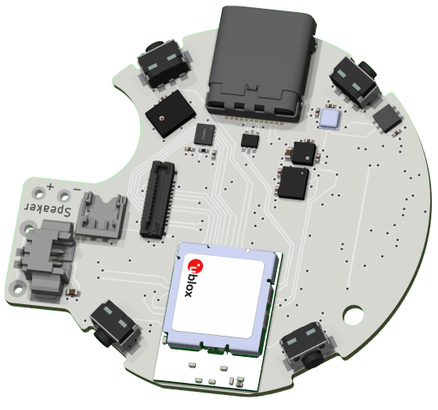
Watch-DevKit-HW

Variant: RELEASED

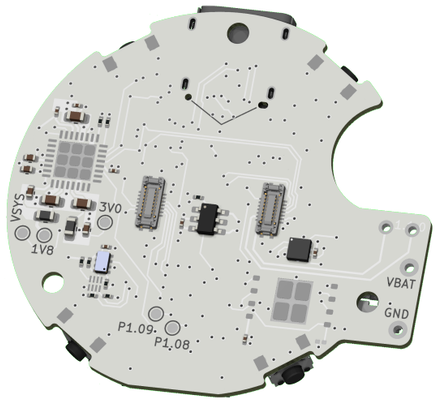
2026-10-16
Rev 1.1.0

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	.....	21	.....	31	.....
2	Block Diagram	12	.....	22	.....	32	.....
3	Power - Topology	13	.....	23	.....	33	.....
4	Project Architecture	14	.....	24	.....	34	.....
5	Power Management	15	.....	25	.....	35	.....
6	Peripherals	16	.....	26	.....	36	.....
7	MCU	17	.....	27	.....	37	.....
8	Extension	18	.....	28	.....	38	.....
9	Revision History	19	.....	29	.....	39	.....
10	.....	20	.....	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as X

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 16-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

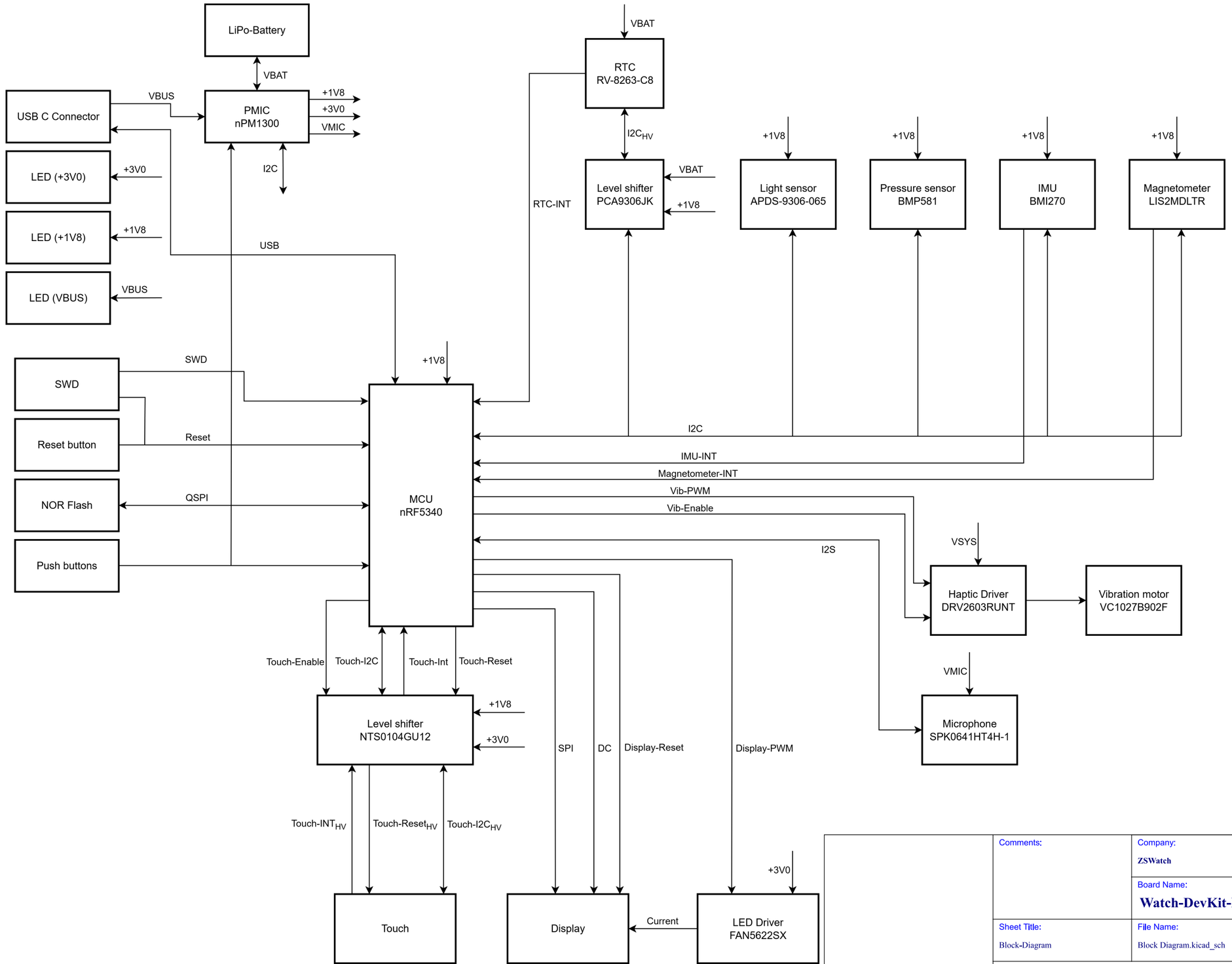
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: ZSWatch		Variant: RELEASED	Git Hash: fccc54e
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
Sheet Title: ZSWatch-Watch	File Name: ZSWatch-Watch.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.1.0
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 9

[2] Block-Diagram



Target specifications:

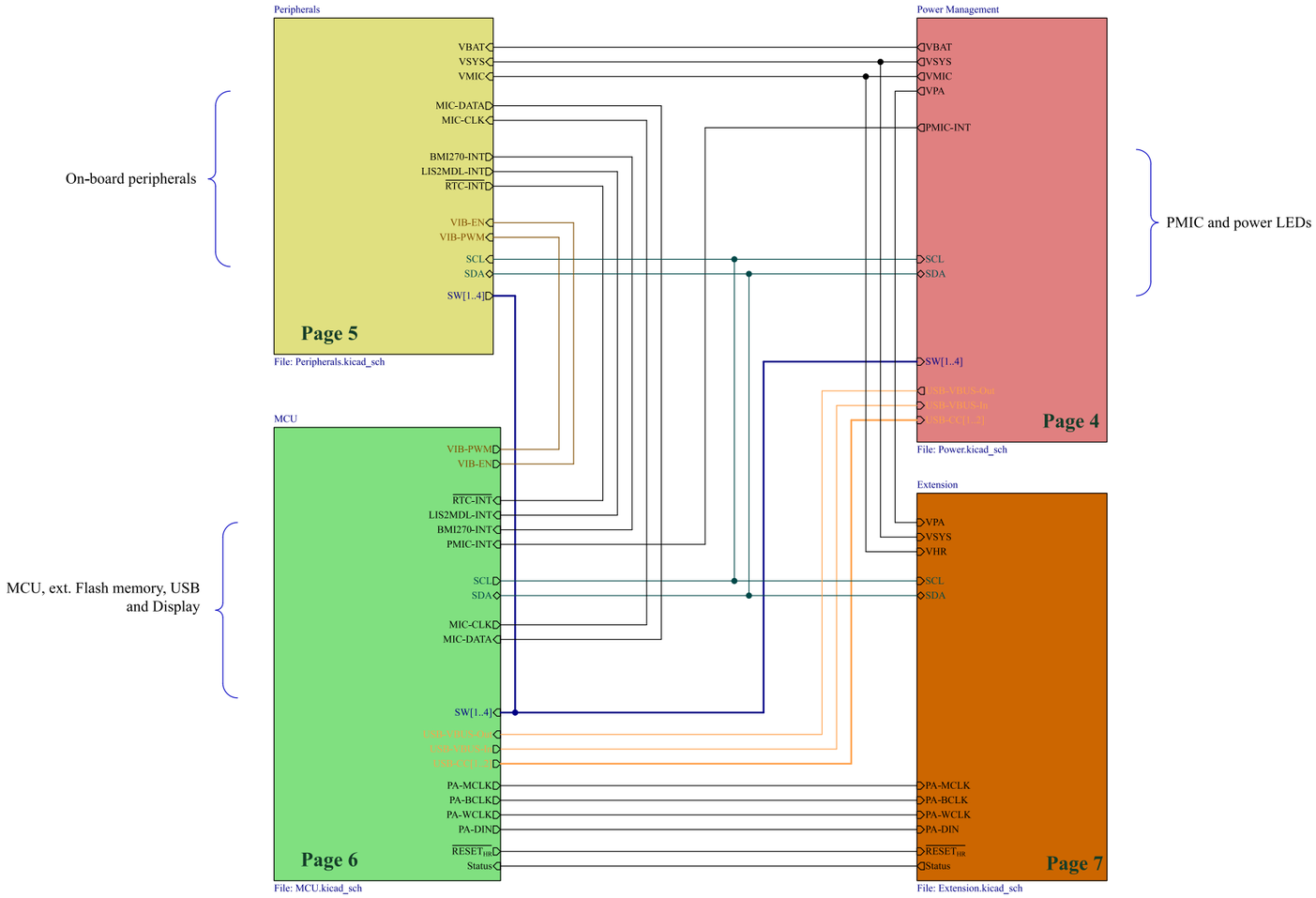
Input voltage:	5 V ± 5%
Battery voltage	3.2 V to 4.2 V
Spec 3	??
Spec 4	??

Comments:	Company: ZSWatch	Variant: RELEASED	Git Hash: fccc54e
	Board Name: Watch-DevKit-HW	Project Name: ZSWatch Watch-DevKit	
Sheet Title: Block-Diagram	File Name: Block Diagram.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13 / Revision: 1.1.0
Sheet Path: /Block Diagram/		Reviewer:	Size: A3 / Sheet: 2 of 9

Target specifications:

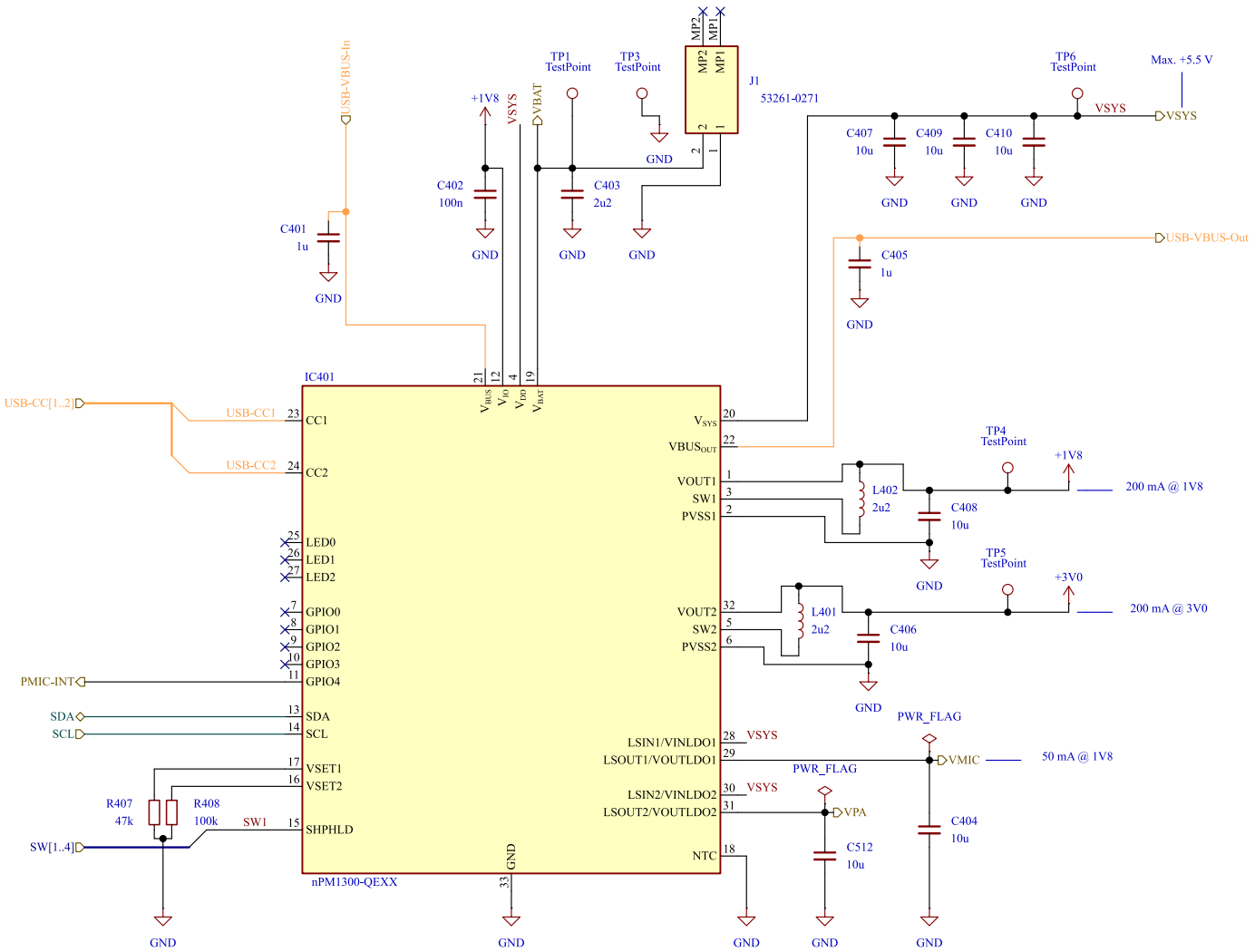
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

Comments:	Company: **ZSWatch**		Variant: RELEASED	Git Hash: fccc54e
	Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**	
Sheet Title:	File Name: Power - Topology.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.1.0
	Sheet Path: /Power - Topology/	Reviewer:	Size: **A3**	Sheet: **2** of **9**



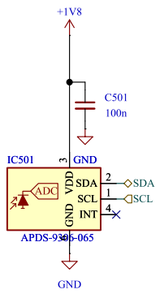
I2C addresses	
BMP581	0x46
RV-8263-C8	0x51
APDS-9306-065	0x52
BMI270	0x68
nPM1300	0x6B
LIS2MDLTR	0x1E

Comments:	Company: ZSWatch		Variant: RELEASED	Git Hash: fccc54e
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
Sheet Title: Power-Management	File Name: Project Architecture.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.1.0
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 9

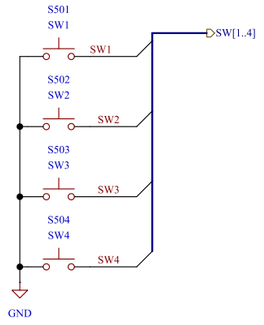


Comments:	Company: **ZSWatch**		Variant: RELEASED	Git Hash: fccc54e
	Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**	
Sheet Title: Power-Management	File Name: Power.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.1.0
Sheet Path: /Project Architecture/Power Management/		Reviewer:	Size: **A4**	Sheet: **4** of **9**

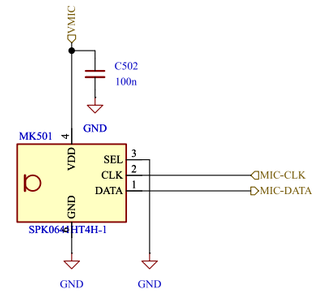
Light sensor



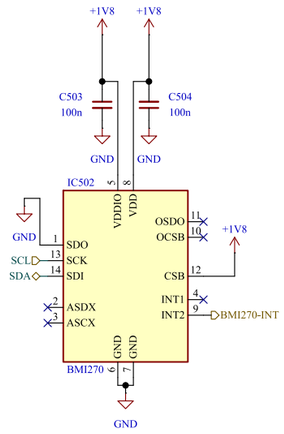
Buttons



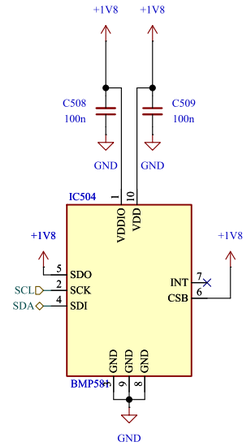
Microphone



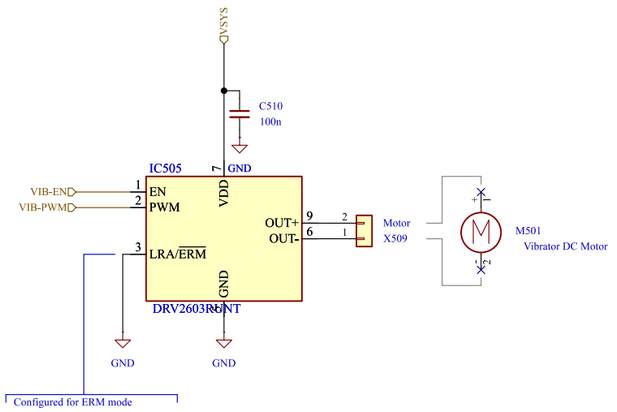
IMU with step counter and gesture detection



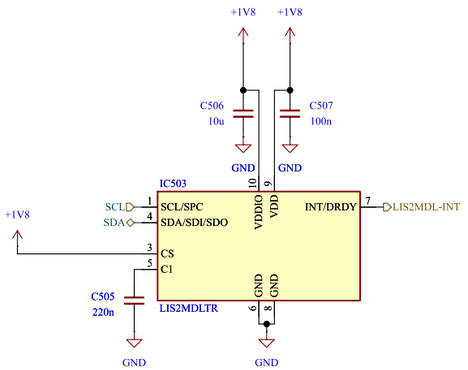
High accuracy pressure sensor



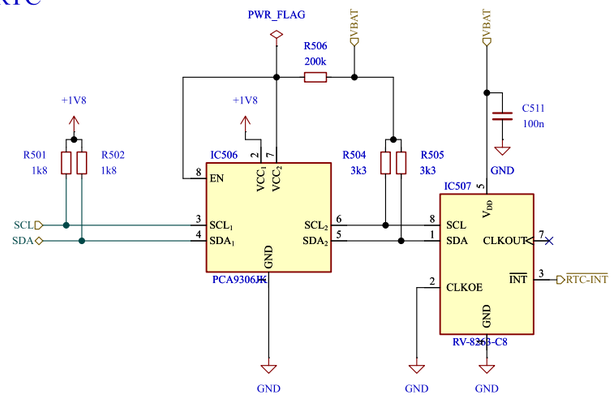
Vibration motor



Magnetometer



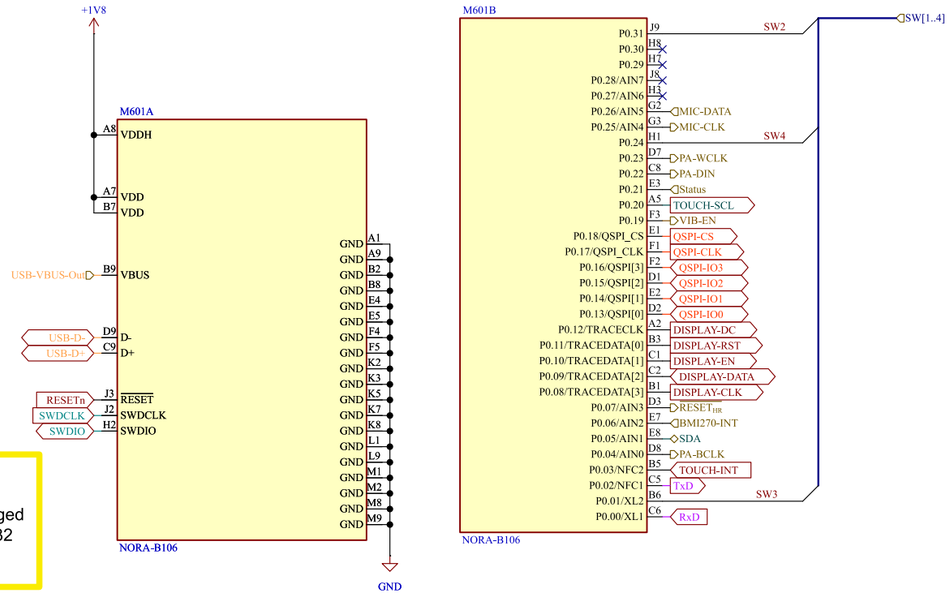
RTC

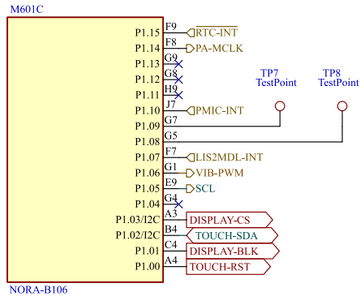


NOTE: CLKOUT is disabled.

Comments:	Company: **ZSWatch**		Variant: RELEASED	Git Hash: fccc54e
	Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**	
Sheet Title: Peripherals	File Name: Peripherals.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.1.0
Sheet Path: /Project Architecture/Peripherals/		Reviewer:	Size: **A3**	Sheet: **5** of **9**

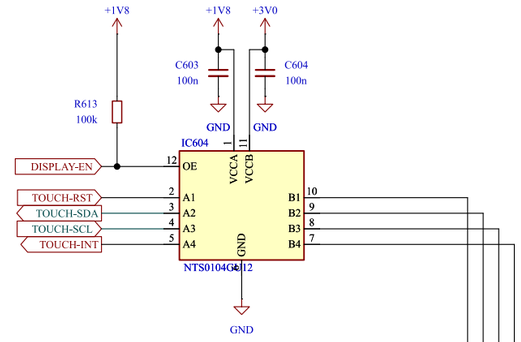
Microcontroller (NORA-B106/nRF5340)

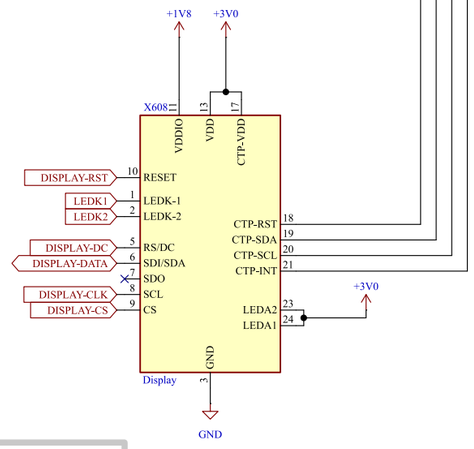




DESIGN NOTE:
The SPI lines for the display can´t be changed and are required for 32 MHz operation.

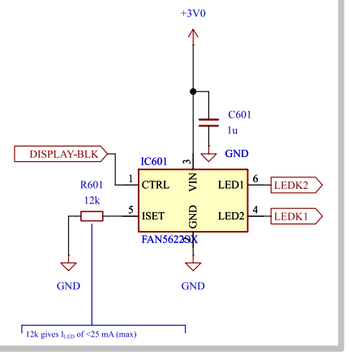
Display



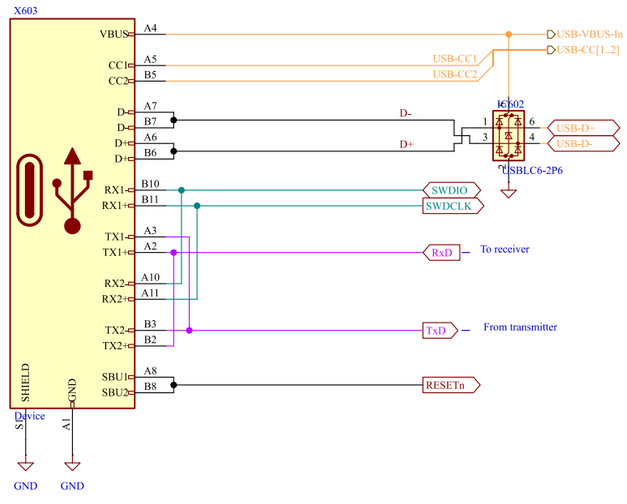


DESIGN NOTE:
NTS0104GU12 has internal pullups.

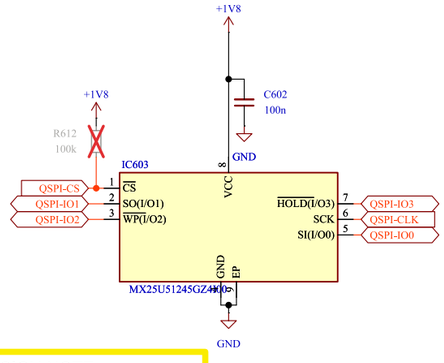
Display backlight



USB

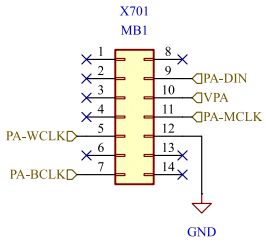


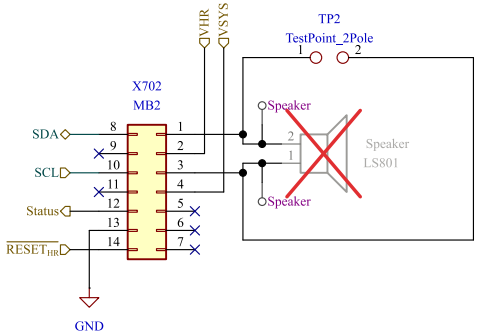
External Flash



DESIGN NOTE:
Make sure that the Flash Memory supports
4k sector erase commands (limited by QSPI driver)!

Comments:	Company: **ZSWatch**		Variant: RELEASED	Git Hash: fccc54e
	Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**	
Sheet Title: MCU	File Name: MCU.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.1.0
Sheet Path: /Project Architecture/MCU/		Reviewer:	Size: **A3**	Sheet: **6** of **9**





	Comments:	Company: **ZSWatch**		Variant: RELEASED	Git Hash: fccc54e
		Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**	
	Sheet Title: Extension	File Name: Extension.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.1.0
	Sheet Path: /Project Architecture/Extension/		Reviewer:	Size: **A4**	Sheet: **7** of **9**

[8] Revision-History

Version 1.0.0 - 2026-02-21

Added:

Initial release

Version 1.1.0 - 2026-10-16

Added

- Test pads for FEM pins, so NORA-W126 (drop-in compatible, with FEM for higher output power) can be mounted instead of NORA-B106 (#21)

- Pads for speaker instead of a speaker connector (#18)

- Extra power test points on the bottom side of the board (#18)

Changed

- Larger 1.25 mm pitch battery connector (Molex 53261-0271) (#18)

- Battery and speaker stickout reduced by 0.5 mm (#18)

- Optional speaker LS801 excluded from the board and BOM (#18)

Comments:	Company: ZSWatch		Variant: RELEASED	Git Hash: fccc54e
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
Sheet Title: Revision-History	File Name: Revision History.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.1.0
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 8 of 9